

Matrices

1) a_{13} .

2) $k=3, p=n$

3) $(AB)^t = B^t A^t$

4) $AB \neq BA$.

5) $\begin{vmatrix} p^5 & q\left(\frac{p^5-1}{p-1}\right) \\ 0 & 1 \end{vmatrix}$

6) $a - iii$
 $b - i$
 $c - iv$
 $d - ii$

7) $a - ii$
 $b - iii$
 $c - iv$
 $d - i$

$$8) a - ii$$

$$b - iv$$

$$c - i$$

$$d - iii$$

$$9) a - iii$$

$$b - iv$$

$$c - i$$

$$d - ii$$

$$10) a - ii$$

$$b - iv$$

$$c - (i)$$

$$d - iii$$

$$11) A^2 = A \cdot A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 9-8 & -6+4 \\ 12-8 & -8+4 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

$$KA = A^2 + 2I.$$

$$KA = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$KA = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$$

$$k \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 3k & -2k \\ 4k & -2k \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$$

$$3k = 3$$

$$\boxed{k = 1}$$

$$12) \quad A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

$$f(A) = A^2 - 4A + 7$$

$$A \cdot A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4-3 & 6+6 \\ -2+2 & -3+4 \end{bmatrix}$$

$$A^2 \Rightarrow \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix}$$

$$\therefore f(A) = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$f(A) = 0$$

Hence Proved

$$D^2 - 4D + 7I$$

$$A^2 = 4A - 7I$$

$$A^3 = A \cdot A^2 = A(4A - 7I)$$

$$A^3 = 4A^2 - 7A$$

$$A^3 = 4(4A - 7I) - 7A$$

$$A^3 = 16A - 28I - 7A = 9A - 28I$$

$$A^4 = A(9A - 28I)$$

$$A^4 = 9A^2 - 28A$$

$$A^4 = 36A - 63I - 28A = 8A - 63I$$

$$A^5 = 8(4A - 7I) - 63A$$

$$A^5 = -31A - 56I$$

$$A^5 = -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} -62 & -93 \\ 31 & -62 \end{bmatrix} - \begin{bmatrix} 56 & 0 \\ 0 & 56 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

$$13) \quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} \quad \begin{matrix} i = 1, 2, 3, 4 \\ j = 1, 2, 3 \end{matrix}$$

$$i = 1$$

$$a_{11} = 2 + 1 = 3$$

$$a_{12} = 2 + \frac{1}{2} = \frac{5}{2}$$

$$a_{13} = 2 + \frac{1}{3} = \frac{7}{3}$$

$$i = 2$$

$$a_{21} = 4 + 2 = 8$$

$$a_{22} = 4 + 1 = 5$$

$$a_{23} = 4 + \frac{2}{3} = \frac{14}{3}$$

$$i = 3$$

$$a_{31} = 6 + 3 = 9$$

$$a_{32} = 6 + \frac{3}{2} = \frac{15}{2}$$

$$a_{33} = 6 + 1 = 7$$

$$i = 4$$

$$a_{41} = 8 + 4 = 12$$

$$a_{42} = 8 + 2 = 10$$

$$a_{43} = 8 + \frac{4}{3} = \frac{28}{3}$$

$$\therefore A = \begin{bmatrix} 3 & \frac{5}{2} & \frac{7}{3} \\ 8 & 5 & \frac{14}{3} \\ 9 & \frac{15}{2} & 7 \\ 12 & 10 & \frac{28}{3} \end{bmatrix}$$

$$15) \begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

$$a-b = -1$$

$$2a+c = 5$$

$$2a-b=0, \quad 2a=b$$

$$3c+d=13$$

$$a-(2a) = -1$$

$$-a = -1$$

$$\boxed{a=1}$$

$$2(1) = b$$

$$\boxed{b=2}$$

$$16) X = \frac{1}{2}(A+A^T)$$

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix}$$

$$(A+A^T)$$

$$\Rightarrow \begin{bmatrix} 6 & 3 & 5 \\ 3 & 8 & 8 \\ 5 & 8 & 16 \end{bmatrix}$$

$$Y = \frac{1}{2}(A-A^T)$$

$$A^T = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 4 & 5 \\ 7 & 3 & 8 \end{bmatrix}$$

$$(A-A^T)$$

$$\begin{bmatrix} 0 & 1 & 9 \\ -1 & 0 & -2 \\ -9 & 2 & 0 \end{bmatrix}$$

$$\Rightarrow X = \frac{1}{2} \begin{bmatrix} 6 & 3 & 5 \\ 3 & 8 & 8 \\ 5 & 8 & 16 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 & 3/2 & 5/2 \\ 3/2 & 4 & 4 \\ 5/2 & 4 & 8 \end{bmatrix} //$$

$$\Rightarrow Y = \frac{1}{2} \begin{bmatrix} 0 & 1 & 9 \\ -1 & 0 & -2 \\ -9 & 2 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & 1/2 & 9/2 \\ -1/2 & 0 & -1 \\ -9/2 & 1 & 0 \end{bmatrix} //$$

$$17) |A| = 5(18+10) + (12-25) + 4(-4-15)$$

$$|A| = 5(28) + (-13) + 4(-19)$$

$$|A| = \overset{140}{\cancel{140}} - 13 - 76$$

$$|A| = \overset{140}{\cancel{140}} - 89$$

$$|A| = \overset{140}{\cancel{140}} - 51 //$$

Cofactors

$$C_{11} = +28$$

$$C_{12} = +13$$

$$C_{13} = -19$$

$$C_{21} = -2$$

$$C_{22} = +10$$

$$C_{23} = +5$$

$$C_{31} = -17$$

$$C_{32} = -13$$

$$C_{33} = +17$$

$$\text{Cofact} = \begin{bmatrix} 28 & 13 & -19 \\ -2 & 10 & 5 \\ -17 & -13 & 17 \end{bmatrix}$$

$$\text{adj } A = (\text{Cofact})^T$$

$$\text{adj } A = \begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -13 \\ -19 & 5 & 17 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$A^{-1} = \frac{1}{51} \begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -13 \\ -19 & 5 & 17 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

$$AX = \textcircled{B}$$

$$X = A^{-1}B$$

$$X = \frac{1}{51} \begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -13 \\ -19 & 5 & 17 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

$$X = \frac{1}{51} \begin{bmatrix} 140 - 4 + 17 \\ 65 + 20 + 13 \\ -95 + 10 - 1 \end{bmatrix}$$

$$X = \frac{1}{51} \begin{bmatrix} 153 \\ 98 \\ -86 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 \\ 98/51 \\ -86/51 \end{bmatrix}$$

$$x = 3$$

$$y = 98/51$$

$$z = -86/51 //$$